KCPC-CS111
Supplementary Reading
Graphs II: Traversal and Shortest Paths

This document contains recommended and further readings for Depth-First Search (DFS), Breadth-First Search (BFS), Dijkstras Algorithm, and 0–1 BFS. These references are intended to deepen conceptual understanding beyond implementation.

I. Core Algorithmic Reading

Competitive Programming Guide (PDF)

`https://duoblogger.github.io/assets/pdf/memonvyftw/guide-t-cp.pdf`

- **Depth-First Search (DFS)** - p. 83
- **Breadth-First Search (BFS)** - p. 85
- **Bellman–Ford** (further reading) - p. 88
- **Dijkstras Algorithm** - p. 89
- **Floyd–Warshall** (further reading) - p. 92

II. 0–1 BFS

CP-Algorithms

`https://cp-algorithms.com/graph/01_bfs.html`

This reference provides a concise explanation of 0–1 BFS, including its correctness argument and time complexity analysis.

III. Formal Textbook Reference

Introduction to Algorithms
(Cormen, Leiserson, Rivest, Stein)

`https://www.cs.mcgill.ca/~akroit/math/compsci/Cormen%20Introduction%20to%20Algorithms.pdf`

- **Priority Queue** - p. 162
- **Breadth-First Search** - p. 594
- **Depth-First Search** - p. 603
- **Bellman–Ford** (further reading) - p. 651

- **Dijkstras Algorithm** - p. 658
- **Floyd–Warshall** (further reading) - p. 693

Recommended Approach:

- Begin with the competitive programming guide for implementation clarity.
- Use CP-Algorithms for structural intuition regarding 0–1 BFS.
- Consult CLRS for formal proofs, invariants, and complexity analysis.